Composite functions

1 Differentiate the following:

a $y = e^x \sin x$

b $y = e^{\sin x}$

c $y = e^x \cos 3x$

d $y = e^{3x} \cos\left(\frac{x}{3}\right)$

e $y = e^{-x} \sin 4x$

f $y = \cos(\ln x)$

g $y = \cos(\ln(-5x))$

h $y = \cos x \ln x$

i $y = e^x \ln x$

j $y = e^{2x} \ln(4x)$

k $y = \log_e(\ln x)$

l $y = \frac{\ln x}{e^{2x}}$

m $y = e^{-2} \ln\left(-6 + x^{-3}\right)$

n $y = \cos\left(\frac{\pi t}{2} - \frac{\pi}{3}\right)$

o $y = x^2 \sin\left(\frac{1}{x}\right)$

p $y = 4\sin\left(\frac{x}{5}\right) - 6e^{2x} + x^{-8}$

q $y = \ln\left(x^2 - 7x - 12\right) - \sqrt{11x}$

r $y = \left(e^{-5x^2} + \cos x\right)^5$

s $y = \frac{e^{-0.2x}}{\sin\left(\frac{\pi}{4}x\right) - x^4}$

t $y = (\cos x + \sin x)e^{6x}$

u $y = \frac{\cos\left(4x - \frac{10\pi}{11}\right)}{(x+3)^2}$

2 Find the derivative of the following:.

a $e^{5x}\cos(x)$

b $x^3 \sin x$

c $\cos^2(x)$

d $\frac{\cos x - \sin x}{\cos x + \sin x}$

e $\frac{4x}{\sin x}$

f $e^{3x}\cos\left(5x + \frac{4\pi}{7}\right)$

g $\frac{\ln x}{\sin x}$

3 For each of the following curves and given points:

i Find an expression for $\frac{dy}{dx}$.

ii Find the exact value of the gradient of the curve at the given point.

a $y = \cos 3x$ at $x = \frac{\pi}{18}$.

b $y = 6 - \cos 3x$ at $x = \frac{\pi}{9}$.

c $y = \sin 4x$ at $x = \frac{\pi}{16}$.

d $y = \sin^2(4x)$ at $x = \frac{\pi}{32}$.

e $y = \cos^2(2x)$ at $x = \frac{\pi}{24}$.

4 Consider the expression $e^{\cos x} \sin(e^x)$.

a If $u = e^{\cos x}$, find $\frac{du}{dx}$.

b If $v = \sin(e^x)$, find $\frac{dv}{dx}$.

c Hence, find the derivative of $y = e^{\cos x} \sin(e^x)$.

5 Consider the function $y = \frac{4x^2 + e^x}{\cos 7x}$.

a If $u = 4x^2 + e^x$, find u'.

b If $v = \cos 7x$, find v'.

c Hence, find y'.

6 Consider the function $y = xe^x$.

a Show that $e^{x + \ln x} = xe^x$.

b Hence, find $\frac{dy}{dx}$, without using the product rule.

7 Determine $\frac{dy}{dx}$, given that $y = u^5$ and $u = \ln(x+5)$.

8 Consider the expression $e^x (1 + \ln 0.3x)^{-2}$.

a If $y = (1 + \ln(0.3x))^{-2}$, find $\frac{dy}{dx}$.

b Hence, find the derivative of $y = e^x (1 + \ln 0.3x)^{-2}$.

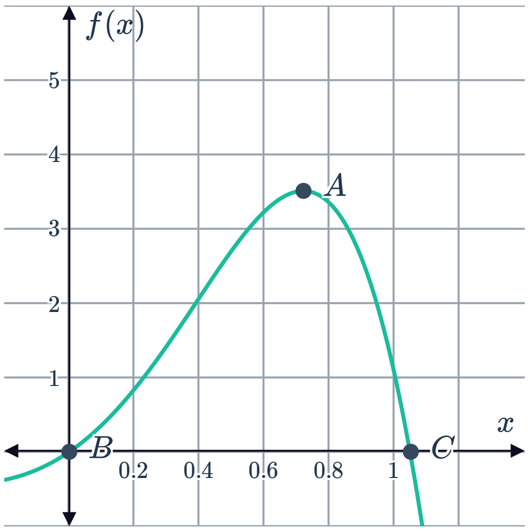
9 Consider the equation $y = \left(2\ln(-3x - x^2) - \cos\left(\frac{\pi}{3}x\right) + \frac{1}{x^4}\right)^3$.

a Find the derivative of $2\ln(-3x - x^2)$.

b Hence differentiate $y = \left(2\ln(-3x - x^2) - \cos\left(\frac{\pi}{3}x\right) + \frac{1}{x^4}\right)^3$.

Stationary points

10 The graph of the function $f(x) = e^{2x} \sin 3x$ is shown:



a Find $f'(x)$.

b Find the x-intercepts, B and C.

c Determine the coordinates of point A, correct to two decimal places.

11 Consider the function $f(x) = x^3 \ln x$, over the domain $e^{-\frac{1}{2}} \leq x \leq e^{-\frac{1}{4}}$.

a Find an expression for $f'(x)$.

b Find the exact values of x such that $f'(x) = 0$.

c Complete the table of values:

x	$e^{-\frac{1}{2}}$	$e^{-\frac{1}{3}}$	$e^{-\frac{1}{4}}$
$f(x)$			

d Hence, state the nature and location of any stationary points.

e Calculate the exact global minimum over the domain.

f Calculate the exact global maximum over the domain.

12 Consider the function $y = \dfrac{\ln 3x}{e^{3x}}$.

- **a** Find the x-intercept of the function.
- **b** Find an expression for $\dfrac{dy}{dx}$.
- **c** Show that the turning point of the graph occurs when $3x \ln 3x - 1 = 0$.
- **d** An approximation to the solution of the equation $3x \ln 3x - 1 = 0$ is $x = 0.58$.
  Complete the table of values, correct to two decimal places.

x	0.01	0.58	1.58
$\dfrac{dy}{dx}$		0	

- **e** Hence, state the nature of the turning point at $x = 0.58$.
- **f** State the coordinates of the turning point, correct to two decimal places.

13 Determine the values of the non-zero constants, a and b, for the following function, given it has a turning point at $(0.25, 1)$:

$$f(x) = axe^{bx}$$

14 Consider the function $f(x) = \ln(\sin 2x)$.

- **a** State the values of x between 0 and 2π for which the function is defined.
- **b** Determine the values of x in the domain $0 \leq x \leq 2\pi$, for which the function has a maximum value.
- **c** State the maximum value of the function.

Applications

15 The curve $y = x \cos x$ passes through the point Q, $\left(\dfrac{\pi}{2}, 0\right)$.
Find the equation of the tangent at point Q.

16 Find the equation of the tangent to the following curves:

- **a** $y = e^x - 3\sin x$ at $x = \dfrac{3\pi}{2}$.
- **b** $y = e^{\cos x}$ at $x = \dfrac{3\pi}{2}$.

17 Researchers have created a model to proj[redacted]e country's population for the next 10 years, where P is the population (in thousands), t years from now. The model is defined by the function:

$$P(t) = \frac{57\,460e^{\frac{t}{7}}}{t+13}$$

- **a** State the current population of the country.
- **b** According to the model, state the current rate of growth of the population, to the nearest thousand.
- **c** Find the rate of population growth 7 months from now, to the nearest thousand.
- **d** Find the rate of population growth 10 years from now, to the nearest thousand.

18 The displacement of a particle moving in rectilinear motion is given by:

$$x(t) = (t-2)^2 + \sin(t-4\pi) + 3t + 25$$

- **a** State the initial displacement of the particle.
- **b** Write an expression for the velocity of the particle, $v(t) = x'(t)$.
- **c** Write an expression for the acceleration of the particle, $a(t) = v'(t)$.

19 A charged particle moves back and forth about the fixed point $x = 0$ (called the origin). Its position, x cm from the origin, after t seconds is given by the equation:

$$x = \sin\left(\pi e^{2t}\right)$$

- **a** Find the particle's initial position.
- **b** Find an expression for the velocity of the particle , $v(t) = x'(t)$.
- **c** Find the exact times for the first and second occasion that the particle comes to a stop.
- **d** Describe the position of the particle when it first comes to a stop, $v(t) = 0$.
- **e** Describe the position of the particle when it comes to a stop for the second time.